Nov 12, '58

Meeting with Don Holmes + group on commercial Applications + needs

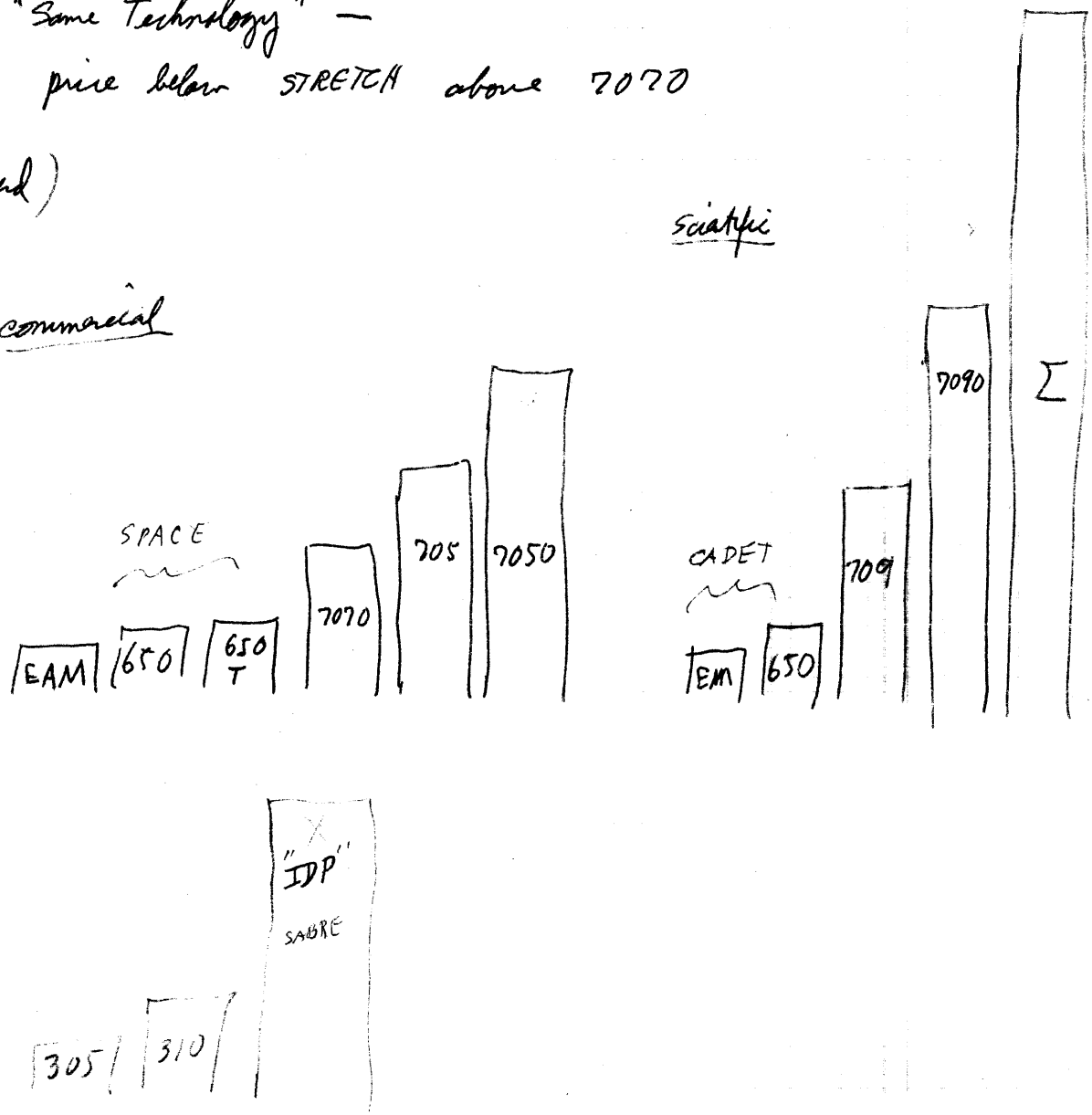
Holmes	Blaauw	Jones
Schmidt	Kolsky	Hank Head
Poland	English	
	England	

Def. of 7000-X :

"Same Technology" —

price below STRETCH above 7070

(Poland)



opinion 2

mkt:

if SPACE, CADET & 7050 } all work

705 customers who want more performance, —

Then there is no space below 2090 + 2050.

competitors:

Many applications require no more performance than 705 but lower price will hurt us. application limited.

7050 is 705 II (or II) program compatible

internal speeds 5X 705 II

((announcement 7050 in next 6 mos — or it will die ?))

2050 market is to present customers only — replacements 75 %

25 % new sales

question: how much batch processing? how much in line?

disk files on 705? being looked into — an SE

Commercial people become application limited much more easily,

explains mkt for 6000 SPACES

600 7070½

& price is more sensitive

more than ½ 705 customers (i.e. 50) can use more power than 705 but maybe not one 7050?

factors { programming costs, politics

{ 100% are still using obsolete programs -- done in 1954 -- could make up & go to four shifts, etc. will this ever be done?

(1) 2 705's needs 2½ 705's could use 1 machine which can do the same job.

(2) 1 705 customer wants to expand application — runs into time limit on combined application (eg payroll, labor distribution, cost accounting)

(3) volume of business increase — workload expansion. eg. population increases (goods + services type customers will need 50% more capacity in Calif by 1965)

(4) Bell Telephone. 125 705-I worth of machine have investigated 7070 — about ½ speed needed. would do job in 8 hrs. — want to do it in 4 hrs.

question: how much would faster tapes help.

729 I — Tape bottleneck

(7070-X / 705 III-TX) 729 III — 2 channels — computer bottleneck
close at 4 channels (250KC)
(can run 16 direct TRC's on 705 II —)

Application area which 705 & 7070 & success will not be able to handle — still batch process but combination

need of { memory, Rams, Tape units } Large prog. storage Large no. of I/O devices

need random access to records (not vsds)

e.g. 1 event in factory may result in 8 records. 4 of which may come back to computer in different programs, – or may go thru entire file to get one record.

Example: motor vehicle departments. – 5 minutes ~~sec~~ search time or less,
injury problem New York, Calif, Penn.
– motor no., registration, name.

a cheap way of doing inquiry
~ 200K transactions a day – half are inquiries

another apple: hasn't been tackled – non-digitally recorded info.
– drawings at least 12 types,
– written documents
– must be recorded, transmitted from place to place, etc.
eg. names of customers – carried along but not processed.

IDP – will be more expensive than batch.
IDP will be used only where one has to do it.
SABRE, CNO, US Steel – are going to 7090 – They consider STRETCH too expensive.

→ 7000-X should go into IDP area, – – –
→ (14 applications were considered at WRL recently 10 were 7090 + two STRETCH)
– 705 couldn't do it – hard to "bolt from outside". (TRC) ← 705 not convenient
– interrupt. ~~705 can be~~ 705 is hard to program.

can people handle an interrupt? TRC is not used by most people – considered too hard.

should be simple to programme

Why is I/O an interleaved package – why doesn't it have its own computer?

Stored program is easy to explain – I/O is bad

Why not line up I/O simply also, – control unit.

– very important from cost to customer point of view.
that this be in hardware.

Compton -- "get" instruction.

costs memory – maybe memory is cheaper.

If machine is above 7090-7050 + below Σ.

1. Read-only memory – key point "cheap" – protected.
2. big memory – cheaper the better –
3. either binary or <u>decimal</u> <u>alphabetic</u> also important – as long as user can use decimal-alph. characters – not need binary
4. variable field extremely important in commercial,
5. grouped records handled simply – end of record gaps etc. are OK if can be done without programer doesn't have to look at seriously – better to not have to look at specially (any record marks is OK)
5. memory to mem. transmission simply done.
6. editing functions – format control,

6. efficient use of hardware
7. Easy I/O control
7. Sort easily — multiprogramed

10. interrupt? — yes for IDP, no otherwise

7. no control panel - rejected by 705 users on 7070
8. SWIFT tapes - lot of them
9. Disks - large more or less random access.

Program compatible with 1401 ? — modularity can build up. — but customer will not be starting from here.

— new applications —

— part time scientific — commercial departments — more & more common

NCR-304X

DATATRON 220

minn. Honeywell Datamatic —

RCA

CDC

TRANSAC

UNIVAC II ? not serious

no real competition in this area — all in IBM — not buying much at all.

5 year plan

'60 8

- 709S
- 7070X
- 705TX

10. want to be able to store instruction counter.

11. basically tape operated machine with SWIFT slow off line peripheral

12. console - desirable.

13. eliminate checkpoint — less checking

- a low speed channel from remote (common-carrier lines) points
- computer-to-computer communication at low speed
- bring in data & batch process.
- multiprog.

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